非线性规划问题（1）-（2）：

We consider the nonlinear programs with the form

$$\text{minimize} \quad (\min) \quad f(x) \tag{1}$$
$$\text{subject to} \quad (\text{s.t.}) \quad h(x) = 0, \quad x \geq 0, \tag{2}$$

where $x \in \Re^n$, $f : \Re^n \to \Re$ and $h : \Re^n \to \Re^m$ are twice continuously differentiable real-valued functions defined on $\Re^n$. If all functions f and h_i $(i = 1, \ldots, m)$ are

对数障碍子问题（4）：

$$\min_x \quad f(x) - \mu \sum_{j=1}^{n} \ln x_j \quad \text{s.t.} \quad h(x) = 0 \tag{4}$$

μ是障碍参数，在子问题（4）或其KKT系统中固定。子问题（4）是等式约束的。采用原始对偶内点法求解（4）存在问题：即使是在良定义的问题上也可能不收敛（jamming difficulty)

子问题（7）-（9）：

For any given parameters $\mu \geq 0$ and $\rho > 0$, and any $x \in \Re^n$ and $s \in \Re^n$, Liu and Dai (2020) defined $z : \Re^{2n} \to \Re^n$, $z = z(x, s; \mu, \rho)$ and $y : \Re^{2n} \to \Re^n$, $y = y(x, s; \mu, \rho)$ by components to be functions on (x, s) as follows,

$$z_j(x_j, s_j; \mu, \rho) \equiv \frac{1}{2\rho}\left(\sqrt{(s_j - \rho x_j)^2 + 4\rho\mu} - (s_j - \rho x_j)\right), \tag{5}$$
$$y_j(x_j, s_j; \mu, \rho) \equiv \frac{1}{2\rho}\left(\sqrt{(s_j - \rho x_j)^2 + 4\rho\mu} + (s_j - \rho x_j)\right), \tag{6}$$

where $j = 1, \ldots, n$, $x \in \Re^n$ and $s \in \Re^n$ are variables[1]. Based on definitions (5) and (6), Liu and Dai (2020) proposed to solve an equivalent positive relaxation problem to the logarithmic-barrier subproblem (4) (see Theorem 2.3 of Liu and Dai 2020) in the form

$$\min_{x,s} \ f(x) - \mu \sum_{j=1}^{n} \ln z_j(x, s; \mu, \rho) \tag{7}$$
$$\text{s.t.} \ h(x) = 0, \tag{8}$$
$$z(x, s; \mu, \rho) - x = 0. \tag{9}$$

Lemma 2.1-2.2：关于子问题（7）-（9）的运算结论

Lemma 2.3：子问题（7）-（9）的KKT条件的等价性

极小极大问题（17）：

Now we consider the relaxation problem (7)–(9). By incorporating the "similar" augmented Lagrangian terms on constraints of (9) into the objective function, and taking the maximum with respect to s, we obtain a particular mini-max problem

$$\min_{x\in\{x\in\Re^n\,|\,h(x)=0\}} \left\{ f(x) + \sum_{j=1}^{n} \max_{s_j\in\Re} G(x_j, s_j; \mu, \rho) \right\}, \tag{17}$$

or its equivalent form

$$\min_{x\in\{x\in\Re^n\,|\,h(x)=0\}} \max_{s\in\Re^n} F(x, s; \mu, \rho),$$

where $F : \Re^{2n} \to \Re$, $F(x, s; \mu, \rho) \equiv f(x) + \sum_{j=1}^{n} G(x_j, s_j; \mu, \rho)$ and $G : \Re \to \Re$,

$$G(x_j, s_j; \mu, \rho) \equiv -\mu \ln z_j(x_j, s_j; \mu, \rho) + s_j(z_j(x_j, s_j; \mu, \rho) - x_j) + \frac{1}{2}\rho|z_j(x_j, s_j; \mu, \rho) - x_j|^2.$$

It should be noticed that the extra two terms $s^T(z(x, s; \mu, \rho) - x) + \frac{1}{2}\rho\|z(x, s; \mu, \rho) - x\|^2$ in $F(x, s; \mu, \rho)$ (comparing to (7)) are not the usual augmented Lagrangian terms, since they definitely use the variables of s of the function z as the estimates of Lagrange multipliers, and take the parameter ρ in z as the penalty parameter. Moreover, the barrier parameter μ is used not only in the logarithmic-barrier terms but also in the other terms.

是对（7）-（9）的进一步改进，把（9）式转化为了两个增广拉格朗日项。

Lemma 2.4：根据Lemma2.1,2.2进一步推出一些运算结论

Theorem 2.5：极小极大问题（17）与对数障碍子问题（4）的等价性

Theorem 2.5 *Let $\mu > 0$ and $\rho > 0$. The following two results can be obtained.*

(1) The pair $(x^*, s^*) \in \Re^n \times \Re^n$ is a local solution of the mini-max problem (17) if and only if $x^* > 0$ is a local solution of the logarithmic-barrier subproblem (4) and $s_j^* = \mu/x_j^*$ for all $j = 1, \ldots, n$.

(2) If $(x^*, s^*) \in \Re^n \times \Re^n$ is a local solution of the mini-max problem (17) and $\nabla h(x^*)$ is of full column rank, then there exists a $\lambda^* \in \Re^m$ such that

$$\nabla f(x^*) - \nabla h(x^*)\lambda^* - s^* = 0, \tag{18}$$

$$h(x^*) = 0, \tag{19}$$

$$z^* - x^* = 0, \tag{20}$$

where $z^* = z(x^*, s^*; \mu, \rho)$. Thus, (x^*, λ^*) is a KKT pair of the logarithmic-barrier subproblem (4).

（18）-（20）是极小极大问题（17）相关的系统，包含对数障碍子问题（4）的KKT系统

Corollary 2.6是对它的总结推论：

Corollary 2.6 *Assume $\mu > 0$ and $\rho > 0$, f and h_i $(i = 1, \ldots, m)$ are linear functions on $\Re^n$. The primal-dual pair (x^*, s^*) is a solution of the mini-max problem (17) if and only if there exists a $\lambda^* \in \Re^m$ such that (x^*, λ^*) is a KKT pair of the logarithmic-barrier subproblem (4).*

这里推导出的系统（18）-（20）的残差函数可以作为对得到的解接近最优解的程度的评估函数。

本文的松弛内点法（IPRM）

主要的公式推导

不在子问题中固定μ，而是让它和子问题的解一同迭代，实现自适应

加入关于μ的方程（24）（它是μ的迭代要逐步接近的条件），得到（18）-（20）式的增广系统（24）-（27）

Instead of solving the subproblem (17) directly, we solve the associated system (18)–(20) and consider the extended system of equations of (18)–(20) in the form

$$\mu = 0, \tag{24}$$
$$\nabla f(x) - \nabla h(x)\lambda - s = 0, \tag{25}$$
$$h(x) = 0, \tag{26}$$
$$z - x = 0, \tag{27}$$

where $z = z(x, s; \mu, \rho)$ and $y = y(x, s; \mu, \rho)$ are functions on x and s defined by (5) and (6). Distinct from our recent work (Dai et al. 2020; Liu and Dai 2020) and many interior-point methods for nonlinear programs, we also take μ as a variable in the system (24)–(27) instead of only a parameter in the system (18)–(20) so that μ is updated with the iteration point. This approach has been used successfully in smoothing New-

文中说明了只要μ归零了，ρ取不同值时得到的x_j, s_j总是符合互补松弛条件，可以组成原始的问题（1）-（2）的KKT triple，理论上ρ不影响结果：

inequalities (see Qi et al. 2000), where μ is a vector of smoothing parameters. Note that, for $j = 1, \ldots, n$,

$$z_j(x_j, s_j; 0, \rho) = \frac{1}{2\rho}(|s_j - \rho x_j| - (s_j - \rho x_j)) = \max\{0, x_j - s_j/\rho\},$$
$$y_j(x_j, s_j; 0, \rho) = \frac{1}{2\rho}(|s_j - \rho x_j| + (s_j - \rho x_j)) = \max\{0, s_j/\rho - x_j\}.$$

Thus, for any $j = 1, \ldots, n$, the equality $z_j = x_j$ implies that one has either $x_j = 0$, $s_j \ge 0$, $\rho y_j = s_j$, or $x_j \ge 0$, $s_j = 0$, $\rho y_j = 0$. Therefore, any $(x^*, \lambda^*, s^*) \in \Re^n \times \Re^m \times \Re^n$ satisfying the extended system of equations (24)–(27) is a KKT triple of the original problem (1)–(2).

写出（18）-（20）式的残差函数：

Denote the residual function of the system (18)–(20) as follows,

$$\phi_{(\mu,\rho)}(x, \lambda, s) = \frac{1}{2}\|\psi_{(\mu,\rho)}(x, \lambda, s)\|^2, \tag{28}$$

where

$$\psi_{(\mu,\rho)}(x,\lambda,s)=\begin{bmatrix}\nabla f(x)-\nabla h(x)\lambda-s\\ h(x)\\ z-x\end{bmatrix}.$$

据此可以把（24）-（27）式重写成（29）-（30）式：

$$\mu+\gamma\phi_{(\mu,\rho)}(x,\lambda,s)=0, \tag{29}$$

$$\psi_{(\mu,\rho)}(x,\lambda,s)=0, \tag{30}$$

改写成（29）式是为了让μ和ϕ在迭代中平衡。

使用线性化的操作得到系统（29）-（30）的牛顿法迭代公式：

Suppose that (x_k,λ_k,s_k) is the current primal and dual iterates, $\mu=\mu_k$ and $\rho=\rho_k$ are current values of the barrier and penalty parameters. Let $r_k^d=\nabla f(x_k)-\nabla h(x_k)\lambda_k-s_k$, $r_k^e=z_k-x_k$, $r_k^h=h(x_k)$, and correspondingly $\psi_{(\mu_k,\rho_k)}(x_k,\lambda_k,s_k)$ be the residual of the equation (30) at iterate k. The Newton's equations corresponding to equations (29)–(30) are as follows,

$$\Delta\mu=-\mu_k+\gamma\phi_{(\mu_k,\rho_k)}(x_k,\lambda_k,s_k), \tag{31}$$

$$\nabla_{(\mu,x,\lambda,s)}\psi_{(\mu_k,\rho_k)}(x_k,\lambda_k,s_k)^T p=-\psi_{(\mu_k,\rho_k)}(x_k,\lambda_k,s_k), \tag{32}$$

where $p=(\Delta\mu,d_x,d_\lambda,d_s)$, (31) is derived from $\nabla_{(\mu,x,\lambda,s)}\phi_{(\mu_k,\rho_k)}(x_k,\lambda_k,s_k)=\nabla_{(\mu,x,\lambda,s)}\psi_{(\mu_k,\rho_k)}(x_k,\lambda_k,s_k)\psi_{(\mu_k,\rho_k)}(x_k,\lambda_k,s_k)$ and (32). Let $\Delta\mu_k$ be calculated by (31), then the second equation (32) can be written in detail as

$$\begin{bmatrix}B_k & -\nabla h(x_k) & -I\\ \nabla h(x_k)^T & 0 & 0\\ (Z_k+Y_k)^{-1}Y_k & 0 & \frac{1}{\rho_k}(Z_k+Y_k)^{-1}Z_k\end{bmatrix}\begin{bmatrix}d_x\\ d_\lambda\\ d_s\end{bmatrix}=\begin{bmatrix}-r_k^d\\ -r_k^h\\ r_k^e+\frac{1}{\rho_k}\Delta\mu_k(Z_k+Y_k)^{-1}e\end{bmatrix}, \tag{33}$$

where the term on the variation $\Delta\mu_k$ of μ is moved to the right-hand-side of the linearized equation.

这个方程组的系数可以对称化：

The preceding system can also be equivalently written as the linear system with a symmetric coefficient matrix in the form

$$\begin{bmatrix} B_k + \rho_k (Z_k + Y_k)^{-1} Y_k & -\nabla h(x_k) & -(Z_k + Y_k)^{-1} Y_k \\ -\nabla h(x_k)^T & 0 & 0 \\ -(Z_k + Y_k)^{-1} Y_k & 0 & -\frac{1}{\rho_k}(Z_k + Y_k)^{-1} Z_k \end{bmatrix} \begin{bmatrix} d_x \\ d_\lambda \\ d_s \end{bmatrix}$$
$$= - \begin{bmatrix} \hat{r}_k^d - \Delta\mu_k (Z_k + Y_k)^{-1} e \\ -r_k^h \\ r_k^e + \frac{1}{\rho_k} \Delta\mu_k (Z_k + Y_k)^{-1} e \end{bmatrix},$$

where B_k is the Hessian of the Lagrangian $L(x, \lambda, s) = f(x) - \lambda^T h(x) - s^T x$ or its approximation at (x_k, λ_k, s_k), $z_k = z(x_k, s_k; \mu_k, \rho_k)$, $y_k = y(x_k, s_k; \mu_k, \rho_k)$, $Z_k = \operatorname{diag}(z_k)$, $Y_k = \operatorname{diag}(y_k)$, $\Delta\mu_k = -\mu_k + \gamma\phi_{(\mu_k,\rho_k)}(x_k, \lambda_k, s_k)$, $\hat{r}_k^d = \nabla f(x_k) - \nabla h(x_k)\lambda_k - \rho_k y_k$.

其中的B_k,∇h还要根据问题的具体条件计算出来。

这里面就可以用（31）和（32）式把μ, d_x, d_λ, d_s的迭代方向求出来。

选择特定的步长，根据迭代方向和步长就可以更新变量：

Our proposed method generates the new value of parameter μ_{k+1} by

$$\mu_{k+1} = (1 - \alpha_k)\mu_k + \gamma\alpha_k\phi_{(\mu_k,\rho_k)}(x_k, \lambda_k, s_k)$$

and the new primal and dual iterates by a line search procedure

$$x_{k+1} = x_k + \alpha_k d_{xk}, \quad \lambda_{k+1} = \lambda_k + \alpha_k d_{\lambda k}, \quad s_{k+1} = x_k + \alpha_k d_{sk},$$

where $(d_{xk}, d_{\lambda k}, d_{sk})$ is the search direction derived from the system (33), and $\alpha_k \in (0, 1]$ is the step-size.

其中，步长通过线搜索得到（用到预定义的参数σ, τ），

μ在初始化时（Step 0.1）和后面每一次迭代中根据上式更新μ_{k+1}之后（Step5.1），要通过特定的方法将其值调整到合适范围内，使用预定义参数ϵ, η, γ_0

惩罚系数ρ使用特定经验公式来更新：

. Update ρ_k to $\rho_{k+1} = \max\{\rho_k, \sigma \|s_{k+1}\|_\infty / \max(\|x_{k+1}\|, 1)\}$

Algorithm 1

Algorithm 1 A novel primal-dual interior-point relaxation method for problem (1)–(2)

Given $(x_0, \lambda_0, s_0) \in \Re^n \times \Re^m \times \Re^n$, $B_0 \in \Re^{n\times n}$, $\mu_0 > 0$, $\rho_0 > 0$, $\eta > 1$, $\gamma_0, \delta, \tau, \sigma \in (0,1)$. Evaluate z_0 and y_0 by (5) and (6), compute $\phi_{(\mu_0,\rho_0)}(x_0, \lambda_0, s_0)$. Given $\epsilon \in (0, \mu_0)$, set $k := 0$. Set $\ell := 0$, $\mu_{k,\ell} = \mu_k$.

Step 0.1 While $\mu_{k,\ell} > \max\{\eta\phi_{(\mu_{k,\ell},\rho_k)}(x_k, \lambda_k, s_k), \epsilon\}$, set $\mu_{k,\ell+1} = \mu_{k,\ell}/\eta$; evaluate z_k and y_k by (5) and (6) with $\mu = \mu_{k,\ell+1}$, compute $\phi_{(\mu_{k,\ell+1},\rho_k)}(x_k, \lambda_k, s_k)$, set $\ell = \ell + 1$, end.

Set $\mu_k = \mu_{k,\ell}$, $\gamma = \min\{\gamma_0, \mu_k/\phi_{(\mu_k,\rho_k)}(x_k, \lambda_k, s_k)\}$.

While $\mu_k \le \epsilon$ and $\phi_{(\mu_k,\rho_k)}(x_k, \lambda_k, s_k) \le \epsilon$, stop the algorithm.

Step 1. Calculate $\Delta\mu_k$ by $\Delta\mu_k = -\mu_k + \gamma\phi_{(\mu_k,\rho_k)}(x_k, \lambda_k, s_k)$.

Step 2. Solve the linear system (33) to obtain $d_k \equiv (d_{xk}, d_{\lambda k}, d_{sk})$.

Step 3. Select the step-size $\alpha_k \in (0, 1]$ to be the maximal number in $\{1, \delta, \delta^2, \ldots\}$ such that the inequality

$$\begin{aligned}\phi_{(\mu_k+\alpha_k\Delta\mu_k,\rho_k)}&(x_k + \alpha_k d_{xk}, \lambda_k + \alpha_k d_{\lambda k}, s_k + \alpha_k d_{sk}) \\ &\le (1 - 2\tau\alpha_k)\phi_{(\mu_k,\rho_k)}(x_k, \lambda_k, s_k)\end{aligned} \tag{34}$$

is satisfied.

Step 4. Set $\mu_{k+1} = \mu_k + \alpha_k\Delta\mu_k$, $x_{k+1} = x_k + \alpha_k d_{xk}$, $s_{k+1} = s_k + \alpha_k d_{sk}$, and $\lambda_{k+1} = \lambda_k + \alpha_k d_{\lambda k}$.

Step 5. Update ρ_k to $\rho_{k+1} = \max\{\rho_k, \sigma\|s_{k+1}\|_\infty/\max(\|x_{k+1}\|, 1)\}$. Evaluate by (5) and (6)

$$z_{k+1} = z(x_{k+1}, s_{k+1}; \mu_{k+1}, \rho_{k+1}) \text{ and } y_{k+1} = y(x_{k+1}, s_{k+1}; \mu_{k+1}, \rho_{k+1}),$$

compute $\phi_{(\mu_{k+1},\rho_{k+1})}(x_{k+1}, \lambda_{k+1}, s_{k+1})$.

Set $\ell := 0$, $\mu_{k+1,\ell} = \mu_{k+1}$.

Step 5.1 While $\mu_{k+1,\ell} > \max\{\eta\phi_{(\mu_{k+1,\ell},\rho_{k+1})}(x_{k+1}, \lambda_{k+1}, s_{k+1}), \epsilon\}$, set $\mu_{k+1,\ell+1} = \mu_{k+1,\ell}/\eta$; evaluate z_{k+1} and y_{k+1} by (5) and (6) with $\mu = \mu_{k+1,\ell+1}$, compute $\phi_{(\mu_{k+1,\ell+1},\rho_{k+1})}(x_{k+1}, \lambda_{k+1}, s_{k+1})$, set $\ell = \ell + 1$, end.

Set $\mu_{k+1} = \mu_{k+1,\ell}$, $\gamma = \min\{\gamma_0, \mu_{k+1}/\phi_{(\mu_{k+1},\rho_{k+1})}(x_{k+1}, \lambda_{k+1}, s_{k+1})\}$.

Step 6. Update B_k to B_{k+1}, set $k := k + 1$.

End (while)

- **输入参数**
- **初始化（求z_0, y_0和残差函数的值，Step 0.1和Set μ_k, γ），与Step 5的部分对应**
- **循环（直到终止条件成立）**
  - **Step 1 计算$\Delta\mu_k$（（31）式）**
  - **Step 2 求解迭代方向d_k（方程组（33））**
  - **Step 3 线搜索α_k**
  - **Step 4 更新μ和迭代点**
  - **Step 5**
    - **更新惩罚系数ρ**
    - **重新求下一次迭代的z, y和残差函数的值**
    - **Step 5.1 循环，调整μ, γ值，同时重新计算z, y, ϕ值**
  - **Step 6 重新计算B_k（用于下一次迭代的系统（33））**

后面的几段有关于调整μ的作用的解释，还有对Algorithm 1的良定义的证明。

二次规划的计算公式

- 二次规划问题：
  - $\min f(x) = \frac{1}{2}x^\top Gx + x^\top c$
  - $h(x) = Ax - b = 0, x \ge 0$
- $L(x, \lambda, s) = f(x) - \lambda^\top h(x) - s^\top x$

- $B_k = diag\{G, O, O\}$
- $\nabla f(x_k) = Gx_k + c$
- $\nabla h(x_k) = A^\top$